Pranav Saxena

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Education

Birla Institute Of Technology And Science, Pilani — Goa, India
Bachelor of Engineering in Electronics and Communication — *Oct 2022 – May 2026*

Experience

Visiting Researcher — Jun 2025 – Present
Carnegie Mellon University — *Pittsburgh, USA*

- Pursued my undergraduate thesis onsite at Carnegie Mellon University's Robotics Institute under the supervision of Dr. Wenshan Wang and Dr. Ji Zhang from Jun 2025–Dec 2025, continuing remotely since Jan 2026.
- Working on 3D Object Representation Learning.
- Previously worked on Vision-Language Navigation for Outdoor Scenarios and co-organized the CMU-VLA Challenge.

Research Intern — May 2024 – Dec 2025
National University of Singapore — *Remote*

- Research intern at MARMoT Lab, NUS under the supervision of Prof. Guillaume Sartoretti.
- Worked on 3D Scene Graph Generation using Vision-Language Models.
- Designed a generalized framework for Open-Vocabulary 3D Gaussian Splatting, eliminating the need for a per-scene autoencoder.
- Developed a Next-Best-View Policy for 3D Reconstruction using Neural Radiance Fields (co-supervised by Prof. Marija Popović).

Publications and Preprints

Pranav Saxena, Nishant Raghuvanshi, Neena Goveas. *"UAV-VLN: End-to-End Vision-Language Guided Navigation for UAVs"*. European Conference on Mobile Robots 2025 (Oral). [Paper]

Pranav Saxena, Avigyan Bhattacharya, Ji Zhang, Wenshan Wang. *"LLM-RG: Referential Grounding in Outdoor Scenarios using Large Language Models"*. Human-Aware Embodied AI Workshop, IROS 2025. [Paper]

Pranav Saxena, Jimmy Chiun. *"ZING-3D: Zero-Shot Incremental 3D Scene Graphs via Vision-Language Models"*. Multimodal Embodied Interaction in Robots Workshop, ICRA 2026. [Paper]

Pranav Saxena. *"Gen-LangSplat: Generalized Language Gaussian Splatting with Pre-Trained Feature Compression"*. preprint. [Paper]

Projects

3D Object Representation Learning — Oct 2025 – Present
Carnegie Mellon University

- Creating a generalized 3D encoder that can be used to represent 3D objects.
- The learned embeddings could be used for various downstream tasks like 3D Object Segmentation, 3D Shape Completion, 3D Captioning.

Vision Language Navigation for Outdoor Scenarios | Paper, Code — Jun 2025 – Present
Carnegie Mellon University

- Developed a zero-shot hybrid referential grounding pipeline for outdoor driving scenarios.
- Leveraged off-the-shelf Vision-Language Models (VLMs) for detailed visual understanding and Large Language Models (LLMs) for symbolic reasoning, without end-to-end retraining.
- Achieved 64.72% and 67.91% accuracy on val and test subsets of the Talk2Car dataset, respectively.
- Currently developing a benchmark dataset to further advance referential grounding and multimodal reasoning in dynamic outdoor scenes.

3D Scene Graph Generation using VLMs | Paper Apr 2025 – Sept 2025

National University of Singapore (NUS)

- Developed a zero-shot pipeline for incremental 3D scene graph generation using Vision-Language Models (VLMs), enabling open-vocabulary reasoning and geometric grounding.
- Achieved 97% node labels accuracy on scenes in the Replica dataset and 96% on scenes in the HM3D dataset.
- Subsequently extended the framework for object-goal navigation tasks.

Efficient 3D Open Vocabulary Gaussian Splatting | Paper, Code Dec 2024 – May 2025

NUS, TU Delft

- Developed a generalized framework for efficient language-grounded 3D Gaussian Splatting eliminating per-scene autoencoder training and improving cross-scene generalization.
- Implemented a cross-scene generalized autoencoder trained on ScanNet for CLIP feature compression into a 16-dimensional latent space.
- Achieved 84.4% localization accuracy on the LERF dataset and 93.3% mIoU on the 3D-OVS dataset, with a nearly 2× reduction in training time.

Next-Best-View Policy for NeRFs | Code May 2024 – Nov 2024

NUS, TU Delft, Purdue University

- Developed an attention-based Next-Best-View (NBV) policy for Neural Radiance Fields to select the most informative viewpoints for 3D reconstruction.
- Implemented a Vision Transformer (ViT) that encodes multi-view image embeddings and candidate poses to predict ΔPSNR gain, guiding view selection.
- Integrated the framework with pixelNeRF to compute ground-truth ΔPSNR via differentiable rendering and supervise policy training under limited measurement budget.

3D Human Pose Estimation using WiFi signals | Report Jan 2025 – Apr 2025

BITS Pilani, Goa

- Developed a WiFi-based 3D human pose estimation pipeline leveraging Channel State Information (CSI).
- Implemented phase sanitization techniques like unwrapping, wavelet denoising, and NLMS filtering to suppress hardware-induced phase noise and multipath distortions.
- Validated the framework on the MM-Fi dataset, achieving improved 3D joint localization and pose fidelity using sanitized CSI signals.

Technical Skills

Languages: Python, C/C++, SQL
Frameworks: ROS, ROS2, PyTorch, TensorFlow, CUDA, Keras, Flask
Tools: Git, Docker, Arduino IDE, RViZ, Gazebo

Course Work

Artificial Intelligence | Applied Stochastic Process | Digital Image Processing | Artificial Intelligence for Robotics | Game Theory and Its Applications | Statistical Inference and Applications | Computer Programming | Linear Algebra | Calculus | Probability & Statistics

Honors and Awards

Indian Olympiad Qualifier in Astronomy May 2022

Qualified Part-1 (NSEA). Among top 264 students in India to be shortlisted for Part-2 (INAO). Received Certificate of Merit from Indian Association of Physics Teachers.

Pre-Regional Mathematics Olympiad Sept 2019

Qualified Pre-Regional Mathematics Olympiad organized by HBCSE. Among the top 300 students in my state to get qualified.

Leadership

Organizer, CMU-VLA Challenge

Lead organizer of the CMU-VLA Challenge 2025, held as part of IROS 2025. Managed the overall organization and evaluation of the challenge.

Founder and Vice-Captain, Team Artemis

Founded Team Artemis, BITS Pilani-Goa's autonomous drone team. Led the automation subdivision. Ranked 3rd out of 100 teams in Phase-I in nationwide competition, SAE Aerothon'24. Won Best Mission Award in final round.